

# MOTOR INSURANCE CLAIM TRIAGE ASSISTANT

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# BUSINESS PROBLEM AND PAIN POINTS

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บริษัทประกันรถยนต์ได้รับ Claim จำนวนมากในแต่ละวัน ปัจจุบัน  
Claim Officer ต้องตรวจสอบข้อมูลหลายส่วนด้วยตนเอง

- Claim Description
- Policy Rules
- Submitted Documents
- Customer Claim History

กระบวนการ Manual Review ทำให้เกิดปัญหาหลักดังนี้



Manual review effort



Inconsistent triage quality



Delayed risk detection



Slow customer response

# TARGET USERS



**Motor Insurance Claim Officer / Claim Agent**

# AI USE CASE

AI Recommendation



Claim Officer Review



Human Final Decision

## Input

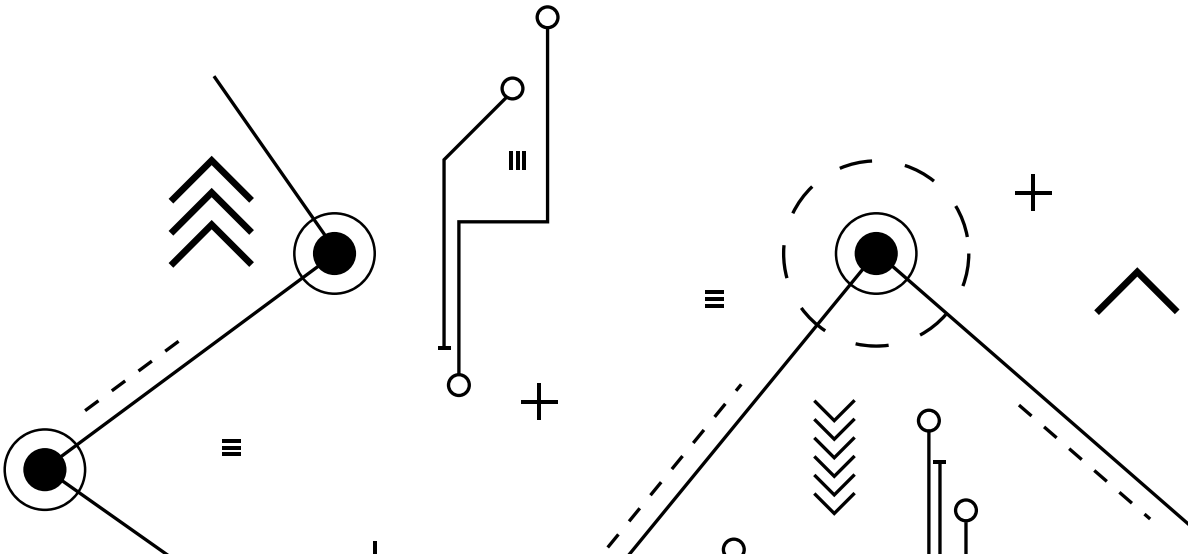
- Claim Description
- Incident Date
- Submitted Documents
- Claim History

## AI Assistant

- Understand & Extract Claim Facts
- Assess Initial Coverage
- Identify Missing Documents
- Detect Risk Flags
- Recommend Routing
- Explain Reasoning

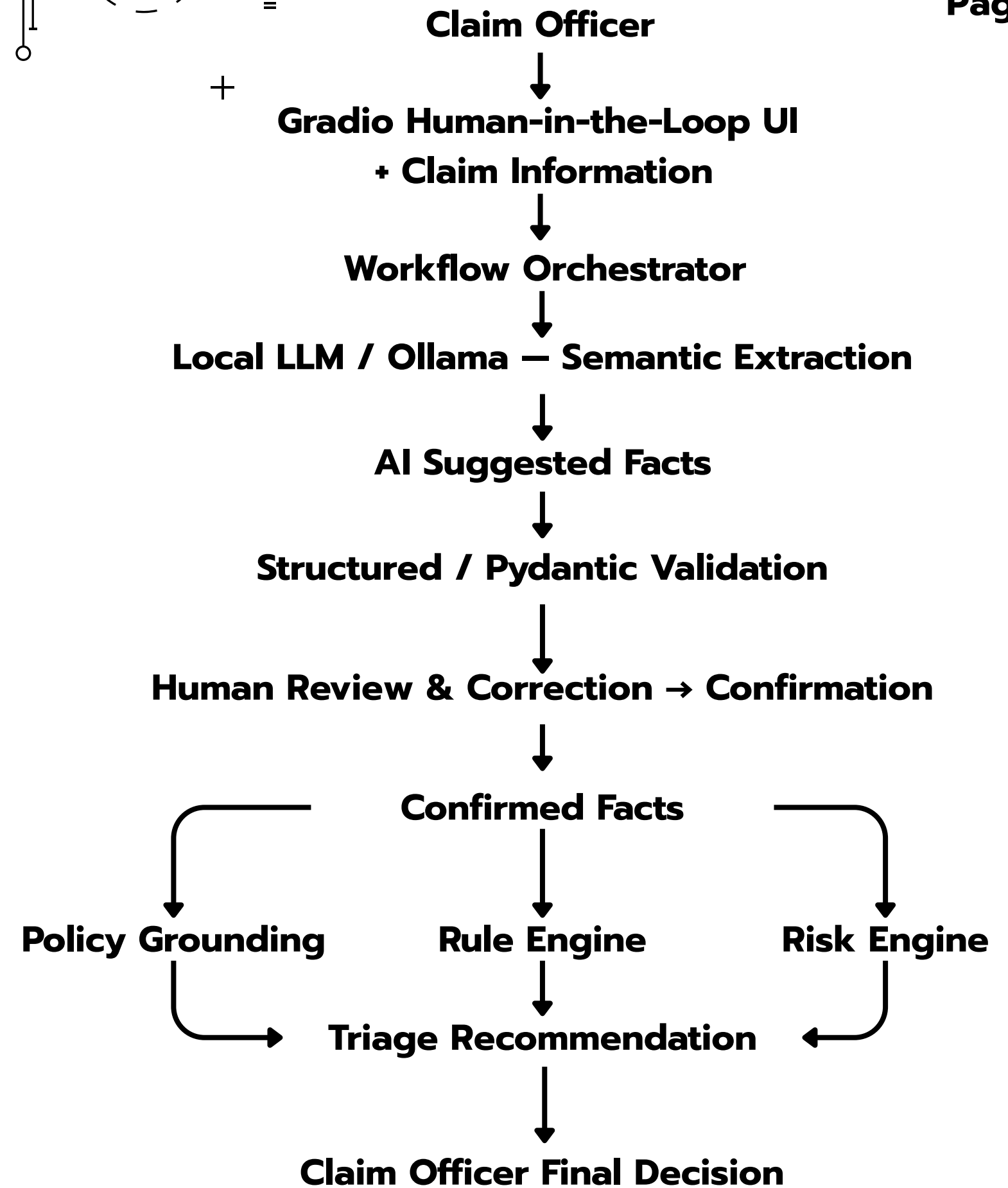
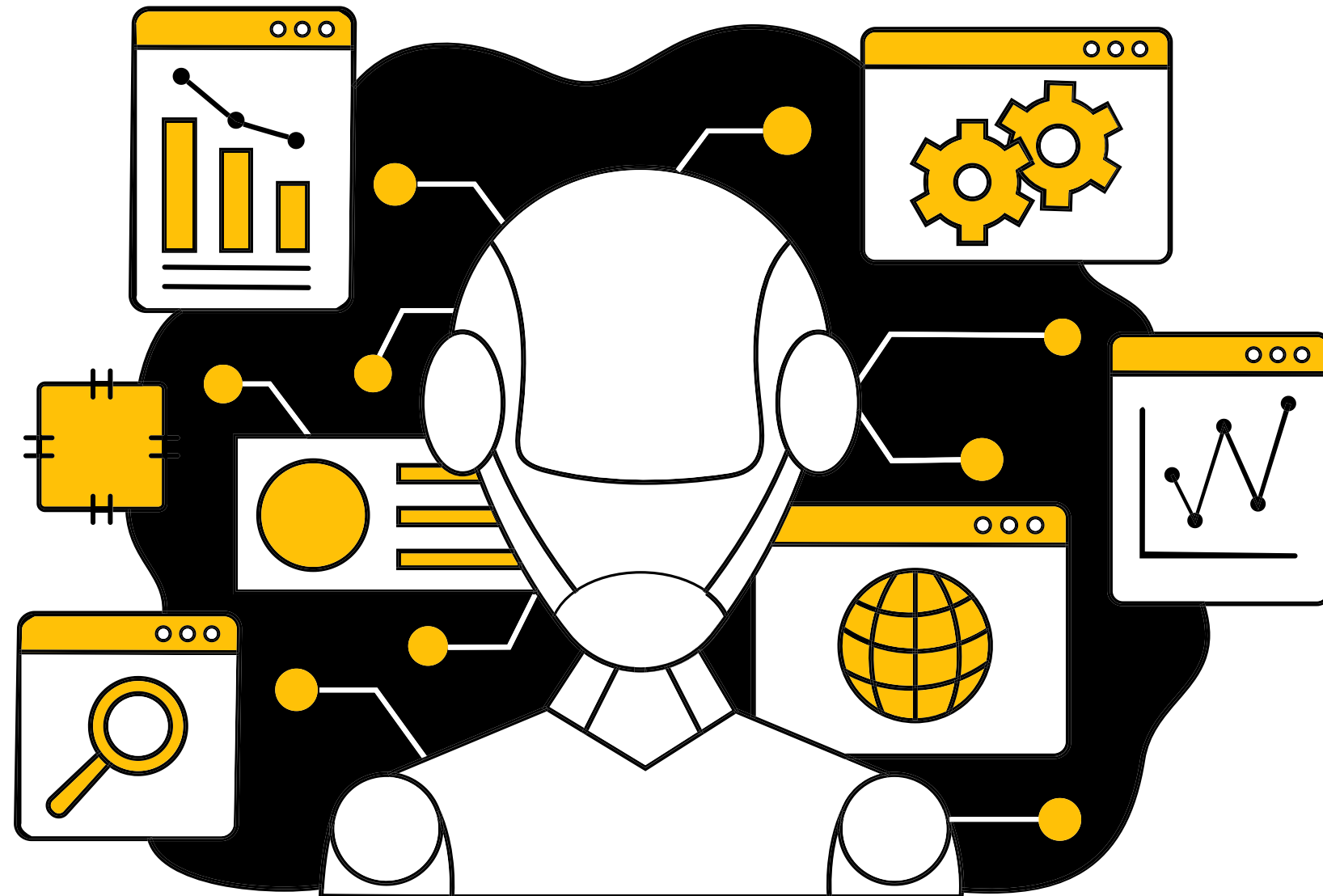
## Output

- Claim Summary
- Coverage Assessment
- Missing Documents
- Risk Flags
- Recommended Routing
- Reasoning
- Confidence

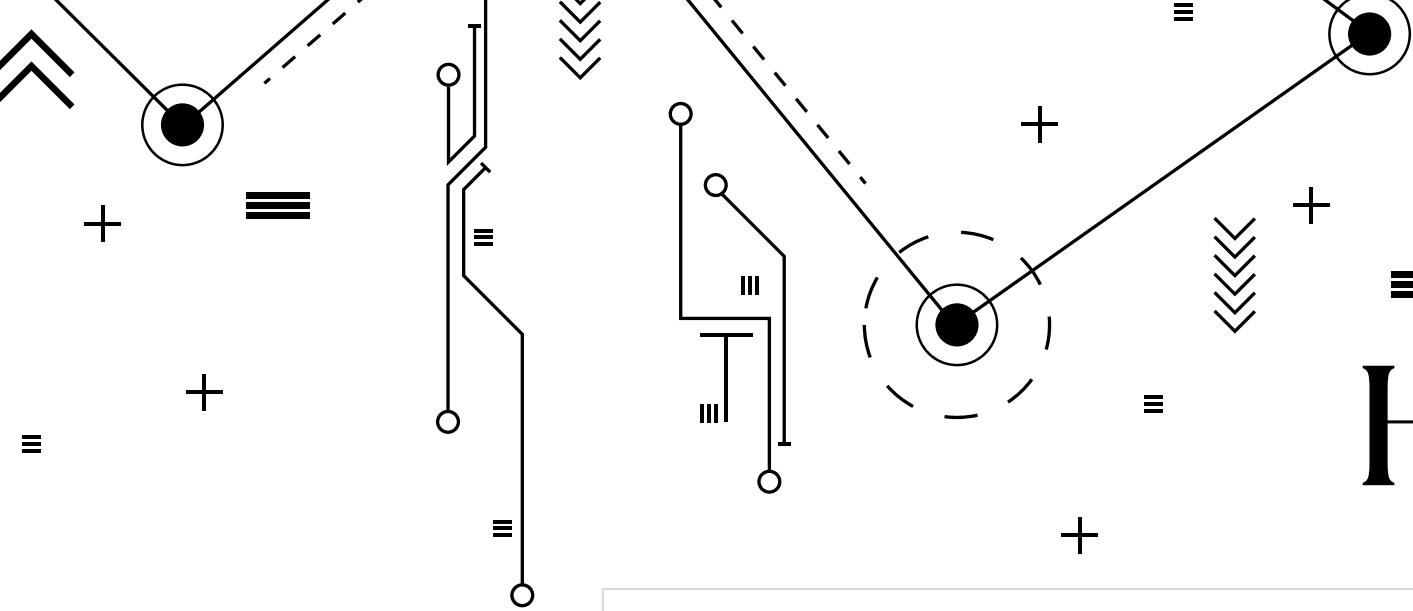


# SOLUTION ARCHITECTURE

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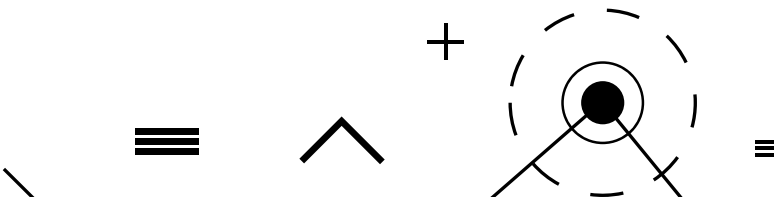






# HOW IT WORK

| Component                       | Responsibility                                                |
|---------------------------------|---------------------------------------------------------------|
| Claim Officer                   | Submit claim information and supporting details               |
| Gradio Frontend                 | Capture user inputs through chatbot and structured forms      |
| Workflow Orchestrator           | Control workflow execution and coordinate downstream services |
| Local LLM / Ollama              | Analyze claim narratives and extract key information          |
| Policy Grounding / RAG          | Retrieve relevant insurance policies and guidelines           |
| Rule Engine                     | Validate claim against predefined business rules              |
| Risk Engine                     | Assess risk level and identify suspicious patterns            |
| Structured Validator (Pydantic) | Validate data structure and output consistency                |
| Triage Recommendation           | Recommend Approve, Review, or Reject decision                 |
| Human Review                    | Conduct final review and make the final claim decision        |



## Motor Insurance Claim Triage Assistant

**Human-in-the-Loop Prototype:** The Local LLM proposes facts only. A Claim Officer must review and confirm/correct them before deterministic triage.

### Step 1 — Claim Information

Demo Case

Assignment Case 3

Claim ID

DEMO-3

Customer

Demo Customer 3

Vehicle

Demo Vehicle 3

Claim Description

Vehicle stolen from condominium parking

Customer Claim History

No prior claim

Incident Date (YYYY-MM-DD)

Claim Submitted Date (YYYY-MM-DD)

Submitted Documents

☒ Claim form

☐ Copy of driving license

☒ Vehicle registration

☐ Photos of damage

☐ Incident report

☐ Police report

☐ Third-party contact information and evidence

☐ One unclear photo

Analyze Claim with AI

# EVALUATION & TESTING

## AI Risk, Governance & Production Readiness Evaluation Areas

- Claim Understanding
  - Coverage Assessment
  - Missing Documents
  - Risk Flags
  - Routing
  - Reasoning
  - Confidence

## Test Result

| Case | Scenario                                | Expected                               | Actual                                 | Result         |
|------|-----------------------------------------|----------------------------------------|----------------------------------------|----------------|
| TC01 | Normal collision                        | Appropriate normal triage              | Manual Review – unresolved information | <b>PARTIAL</b> |
| TC02 | Illegal street racing                   | Not Covered → Rejection Review         | Not Covered → Rejection Review         | <b>PASS</b>    |
| TC03 | Vehicle theft                           | Police Report required → Manual Review | Missing Police Report → Manual Review  | <b>PASS</b>    |
| TC04 | Severe damage + weak evidence           | Fraud Review                           | Fraud Review                           | <b>PASS</b>    |
| TC05 | Late submission >30 days without reason | Manual Review                          | Manual Review                          | <b>PASS</b>    |

## Testing Strategy

Deterministic Unit Tests + LLM / End-to-End Evaluation



# AI RISK, GOVERNANCE & PRODUCTION READINESS

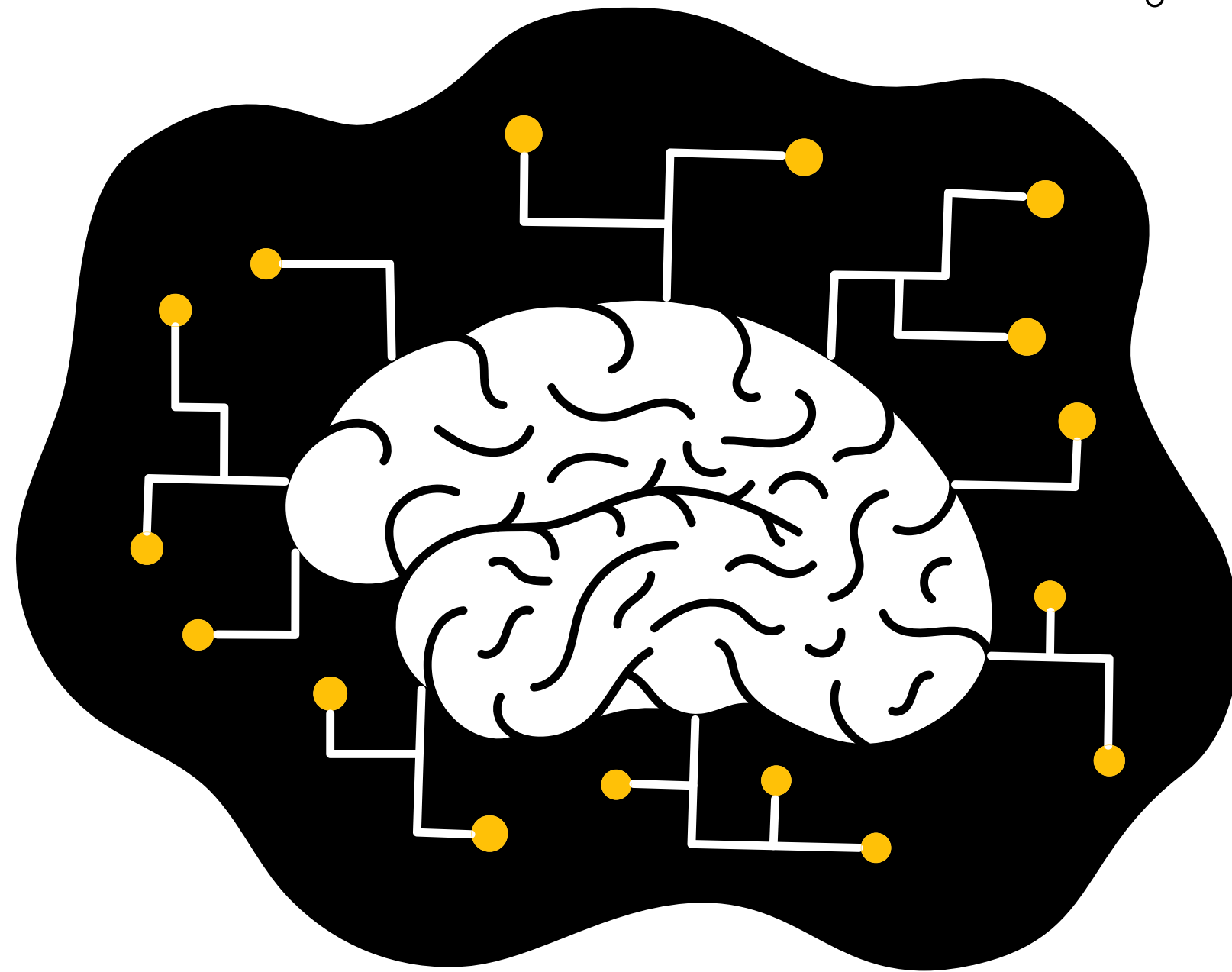
| Risk                | Mitigation                                                                        |
|---------------------|-----------------------------------------------------------------------------------|
| Hallucination       | Policy Grounding (RAG) + Business Rules Validation                                |
| Privacy             | Local LLM deployment, Synthetic Data usage, and PII minimization                  |
| Prompt Injection    | Treat all claim inputs as untrusted data and apply input validation               |
| Inconsistent Output | Structured schema validation using Pydantic                                       |
| Overconfidence      | Confidence scoring combined with Human Review                                     |
| Explainability      | Provide reasoning and references to relevant policies and rules                   |
| Auditability        | Log model version, prompts, retrieved policies, rules executed, and final outputs |

## Human-in-the-Loop

AI Recommends → Human Reviews → Human Decides

## Future

OCR → Real Policy RAG → Core Insurance Integration  
→ Monitoring & Audit



# SUMMARY

Use AI where AI is strong,  
use deterministic rules where  
precision matters,  
and keep humans accountable for  
the final claim decision.

